CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

11.1 Atoms, Nuclei & Radiation

Syllabus Section: 11. Particle Physics **Paper Type:** Multiple Choice (Paper 1) **Total:** 35 Questions (35 Marks)

Question 1 Easy [1 mark]

A Geiger-muller tube held at a distance of about 12 cm from a radioactive source. The radiation emitted from the nuclear isotope was entirely stopped by a 2 mm thick sheet of lead.

Which of the statements can be deduced from the above information about the emission from the isotope?

A it could be alpha, beta and gamma radiation

B it could be beta and gamma radiation, but not alpha radiation

C it could be alpha and gamma radiation, but not beta radiation

D it could be alpha and beta radiation, but not gamma radiation

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	−1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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- Alpha and beta radiation would be entirely stopped by a few mm of lead
- The emission of gamma radiation would only be reduced as some would still be detected through 2 mm of lead
    - Therefore, option **D** is correct

Question 2 Easy [1 mark]

When β^- particles are emitted from identical nuclei in a substance, which of the following remain constant?

A proton number of the nuclei

B nucleon number of the nuclei

C neutron number of the nuclei

D energy of the β-particles

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Nuclei are made up of nucleons; protons and neutrons
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino $\bar{\nu}_e$
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$
  - The mass is not changed, but the proton number increases by one, the nucleon number stays the same

Question 3 Easy [1 mark]

When aluminium $^{27}_{13}Al$ was bombarded with α-particles the first radioactive substance was made. The product of this reaction was an unstable isotope of phosphorus, $^{30}_{15}P$.

Which of the following is a by-product of this reaction?

A a neutron

B a γ-ray

C a β-particle

D an α-particle

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- An α-particle is made of two protons and two neutrons, $^{4}_{2}\alpha$
- When the $^{27}_{13}Al$ is bombarded with these particles the nuclei increases in mass and proton number as shown in the equation
  - $^{27}_{13}Al \ + \ ^{4}_{2}\alpha \ \rightarrow \ ^{30}_{15}P \ + \ ^{Z}_{A}X$
- 27 + 4 = 30 + Z
  - So, Z = 1
- 13 + 2 = 15 + A
  - So, A = 0
- Therefore, $^{1}_{0}X$ is a neutron

Question 4 Easy [1 mark]

The atom $^{133}_{55}Cs$ is a neutral atom

How many protons, neutrons and electrons are in this atom?

	Protons	Neutrons	Electrons
A	133	55	133
B	55	78	55
C	78	55	78
D	55	133	55

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The symbol $^{133}_{55}Cs$ give the mass number of 133 and an atomic number of 55
  - So, the nucleus contains 55 protons
- As the atom is neutral the number of electrons will be 55
- The mass of the atom is calculated by adding the number of neutrons to protons
  - 133 – 55 = 78 neutrons

Question 5 Easy [1 mark]

The following statements are about α-particles.

Which is correct?

A α-particles emitted from a single radioactive isotope have a continuous distribution of energies

B α-particles have less ionising power than β-particles

C the speeds of α-particles can be as high as 1.5×10^{7} m s^{-1}

D the charge of an α-particle is $+1.60 \times 10^{-19}$ C

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- α-particles have a slower speed than that of gamma or beta due to its mass
  - The typical speed of an α-particle is 10^{6} m s^{-1} to 10^{7} m s^{-1}

A is incorrect as the energy of an α-particle changes as the particle moves away from the nucleus it was emitted from

B is incorrect as α-particles have a higher ionising power than β-particles

D is incorrect as the charge on an α-particle is +2e, a charge of 3.2×10^{-19}

Question 6 Easy [1 mark]

Which of the following statements about the nucleus of the atom is correct?

A nucleon numbers are unchanged by the emission of β-particles

B in some nuclear processes, mass-energy is not conserved

C for some nuclei, the nucleon number can be less than the proton number

D different isotopic nuclei have different proton numbers

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino $\bar{\nu}_e$
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one
- Therefore, nucleon numbers are unchanged by the emission of β-particles

B is incorrect as in a nuclear reaction, the total energy is conserved. The apparent loss of mass is conserved in kinetic energy

C is incorrect as an atom's mass number can never be smaller than its atomic number

D is incorrect as if the atom has a different number of protons, then it is a different atom

Question 7 Easy [1 mark]

When an α-particle is produced by $^{258}_{92}U$ it produces a new atom X.

What are the values of the proton number and nucleon number for atom X?

	Protons	Nucleons
A	92	256
B	90	254
C	92	250
D	88	254

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An α-particle is made of two protons and two neutrons, $^{4}_{2}\alpha$
  - $^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
  - $^{258}_{92}U \rightarrow {}^{254}_{90}X + {}^{4}_{2}\alpha$
- So, the proton number decreases by two
  - Therefore, atom X has 90 protons
- So, nucleon number will reduce by four
  - Therefore, atom X has a nucleon number of 254

Question 8 Easy [1 mark]

A nucleus Q decays into a nucleus R by emitting an alpha particle followed by two beta particles.

Which statement about this nuclear decay is correct?

A nucleus R is an isotope of nucleus Q

B nucleus R has the same nucleon number as nucleus Q

C beta particle decay occurs when a proton changes into a neutron

D the total mass of the products is equal to the mass of the initial nucleus Q

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino $\bar{\nu}_e$
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$
    - The mass is not changed, but the proton number increases by one
- The original nucleus Q will contain ${}^{Z}_{A}X$ protons and neutrons
  - $Q \rightarrow R + {}^{4}_{2}\alpha + 2\,{}^{0}_{-1}\beta$
- Isotopes are nuclei with the same proton number and a different number of neutrons and therefore mass
  - So, alpha decay will decrease the proton number by 2
- Then, beta decay will turn two neutrons into two protons and two beta particles
  - There is no change to the proton number, so, R is an isotope of Q

B is incorrect as there will be a change in the nucleon number as the nucleon will have decreased by four in R.

C is incorrect as beta decay occurs when a neutron is changed to a proton

D is incorrect as there will be a mass defect which results in the release of energy so the total mass will not be conserved

Question 9 Easy [1 mark]

Four nuclei are represented below

$$^{24}_{13}Q \qquad ^{28}_{14}E \qquad ^{25}_{15}G \qquad ^{25}_{12}M$$

Which of the following statements about these nuclei is correct?

A an uncharged atom of element Q has 24 orbital electrons

B when E absorbs a neutron and then emits an alpha particle, nucleus E transforms into M

C nuclei G and M are isotopes of the same element

D nucleus M could transform into Q by emitting a beta particle

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- When E absorbs a neutron the new isotope is $^{29}_{14}E$
- An α-particle is made of two protons and two neutrons, $^{4}_{2}\alpha$
  - $^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + ^{4}_{2}\alpha$
- When the new isotope emits an alpha particle, the nucleon number is reduced by 4
  - So, the new nucleon number is 29 – 4 = 25
- The proton number is reduced by 2
  - So, the new proton number is 14 – 2 = 12
- Therefore, the new atom $^{25}_{12}M$ is nuclei M

A is incorrect as 24 is the nucleon or mass of atom Q, the number of electrons in an uncharged atom of Q will be 13

C is incorrect as nuclei G and M have different proton numbers making them different atoms not isotopes

D is incorrect as in β^- decay, a neutron decays to become a proton and emits a β^- particle and an anti-electron neutrino. The mass is not changed, but the proton number increases by one. Nucleus M after β^- decay would be $^{25}_{13}X$

Question 10 Easy [1 mark]

Plutonium-239 $^{239}_{94}Pu$ decays by emitting α-radiation.

Which nuclide is formed from one of these decay reactions? (The product nuclides are represented by X.)

A $^{239}_{95}X$

B $^{237}_{92}X$

C $^{239}_{93}X$

D $^{235}_{92}X$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- An α-particle is made of two protons and two neutrons, $^{4}_{2}\alpha$
  - $^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4 making this 235
- The proton number is reduced by 2 making this 92
  - $^{239}_{94}Pu \rightarrow {}^{235}_{92}X + {}^{4}_{2}\alpha$

Question 11 Easy [1 mark]

One of the rows in the table shows a pair of nuclei that are isotopes of one another.

Which row is it?

	Nucleon number	Number of neutrons
A	186 180	112 118
B	184 187	110 110
C	186 182	112 108
D	186 186	110 112

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Isotopes are nuclei with the same proton number and a different number of neutrons and therefore mass
- The first nucleus has a nucleon number of 186 and 112 neutrons, so, this will make a total of 186 – 112 = 74 protons
- The second nucleus has a nucleon number of 182 and 108 neutrons, so, this will make a total of 182 – 108 = 74 protons
    - They have the same number of protons, therefore, they are isotopes of each other

A is incorrect as	186 – 112 = 74 180 – 118 = 62

	these nuclei are not isotopes of each other
B is incorrect as	184 – 110 = 74 187 – 110 = 77 these nuclei are not isotopes of each other
D is incorrect as	186 – 110 = 76 186 – 112 = 74 these nuclei are not isotopes of each other

Question 12 Easy [1 mark]

Which statement about alpha, beta and gamma radiation is correct?

A gamma radiation has the greatest ionising power

B beta radiation has the greatest ionising power

C alpha radiation has the greatest ionising power

D alpha, beta and gamma radiation have nearly equal ionising powers

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Alpha, beta and gamma have different properties as shown in the table below
- Alpha radiation has the greatest ionising power

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	−1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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Question 13 Easy [1 mark]

Alpha, beta and gamma radiations have various depths of penetration in matter and different charges.

Which row in the table best shows the penetration and charge of each radiation?

	Alpha	Beta	Gamma
A	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium no charge	not fully absorbed by several mm of lead no charge
B	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead negative charge	absorbed by a sheet of card no charge
C	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead no charge
D	absorbed by a sheet of card positive charge	absorbed by several mm of aluminium negative charge	not fully absorbed by several mm of lead no charge

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	−1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - It is absorbed by card and has a positive charge
- A β^- particle is made of an electron, ${}^{0}_{-1}\beta$
  - It is absorbed by aluminium and is negatively charged
- A γ-ray is an electromagnetic wave
  - It is not fully absorbed by lead and has no charge

Question 14 Easy [1 mark]

Most α-particles pass straight through when fired at a thin sheet of metal foil, but a few are deflected by a large angle.

Which change would increase the **proportion** of α-particles deflected by a large angle?

A using a double thickness foil

B using a foil made of a metal with fewer protons in its nuclei

C using α-particles with greater kinetic energy

D using an alpha source with a higher activity

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The gold foil experiment directed parallel beams of α-particles at a piece of gold foil only 10^{-6} m thick
- The observations made in the experiment were as follows:
  - Most of the α-particles went straight through the foil
  - Some of the α-particles were deflected slightly
  - Some, about 1 in 20 000, were deflected by more than 90˚, appearing to bounce back off the foil
- In order to increase the proportion of α-particles that are deflected by a larger angle, more nuclei would need to present
  - Thicker foil would mean that more nuclei are present

B is incorrect as fewer particles in the nuclei would have less of a deflection

C is incorrect as the number of nuclei is not changing so the kinetic energy of the α-particle will have no effect

D is incorrect as the number of nuclei is not changing, so the higher activity of the α-particle will have no effect.

Question 15 Easy [1 mark]

New insights were discovered when α-particles were fired at a sheet of gold foil.

Which conclusion can be drawn from the results of the experiment?

A the atomic nucleus contains protons and neutrons

B electrons orbit the atomic nucleus

C some atoms of the same element contain different numbers of neutrons

D atomic nuclei occupy a very small fraction of the volume of an atom

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The gold foil experiment directed parallel beams of α-particles at a piece of gold foil only 10^{-6} m thick
- The observations were as follows:
  - Most of the α-particles went straight through the foil
  - Some of the α-particles were deflected slightly
  - Some, about 1 in 20 000, were deflected by more than 90˚, appearing to bounce back off the foil
- These results confirmed that the atom was made up of empty space and that the nucleus only occupied a very small fraction of the atom

A is incorrect as the results obtained do not confirm the presence of both protons and neutrons

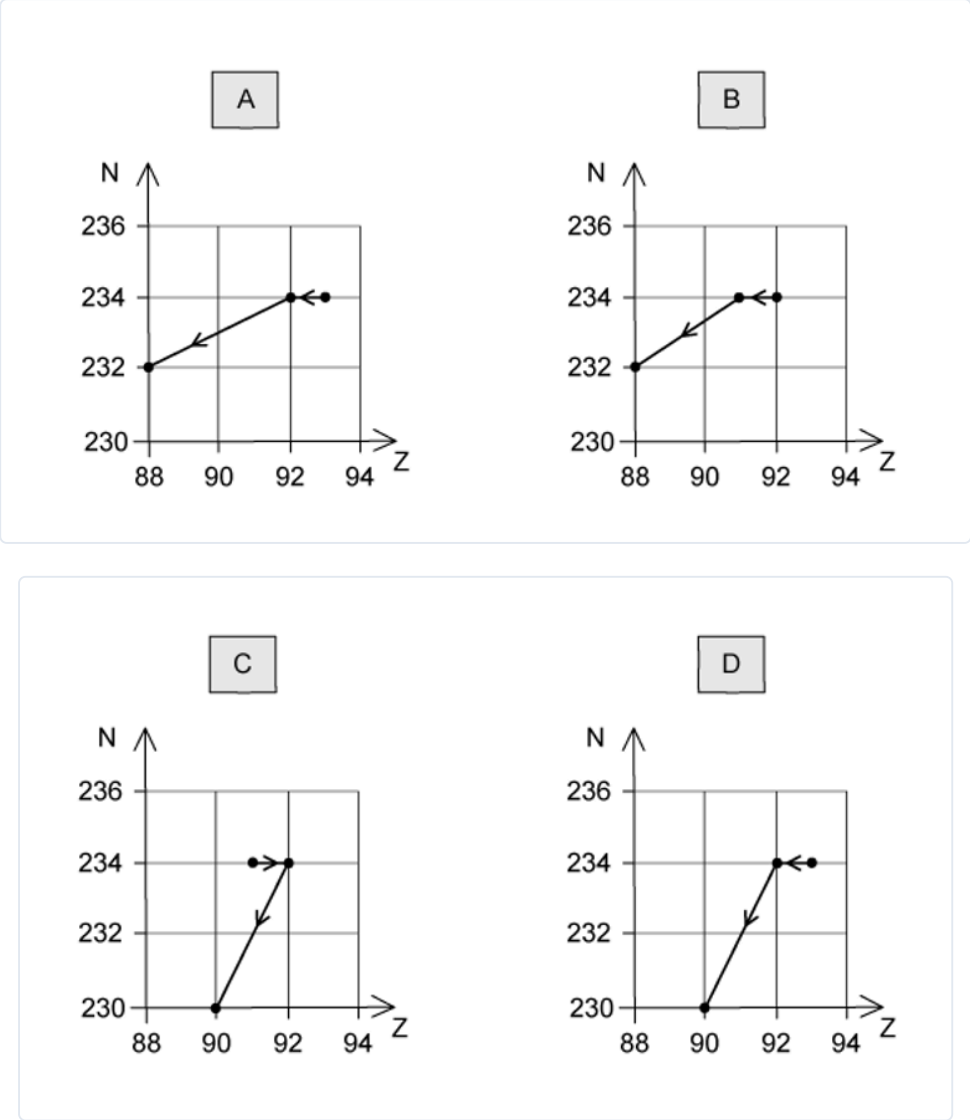
B is incorrect as Bohr first proposed the presence of electrons in orbit in 1913

C is incorrect as the evidence from the gold foil experiment did not support the presence of **isotopes**

Question 16 Medium [1 mark]

A radioactive nucleus is formed by β-decay. This nucleus then decays by α-emission.

The graphs below show the nucleon number N plotted against proton number Z. Which one shows the β-decay followed by the α-emission?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino $\bar{\nu}_e$
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \nu_e$
  - The mass is not changed, but the proton number increases by one; so the Z number has increased from 91 to 92
- An α-particle is made of two protons and two neutrons,
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4 making this 230
- The proton number is reduced by 2 making this 90
  - Graph **C** is therefore the correct option

Question 17 Medium [1 mark]

Nucleus X decays in two stages to produce nucleus Y.

Which decay sequence will result in the highest number of neutrons in nucleus Y ?

A an α-particle followed by a β-particle

B a β-particle followed by a γ-ray

C a β-particle followed by another β-particle

D an α-particle followed by a γ-ray

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- An γ-ray is usually emitted after alpha or beta decay to release excess energy from the nuclei
- The nucleon number is not changed in this case
  - The option that will result in the highest number of neutrons remaining in nucleus Y is therefore **B**

Question 18 Medium [1 mark]

The isotope $^{222}_{86}Rn$ decays in a sequence of emissions to form the isotope $^{206}_{82}Pb$.

It will either emit an α-particle or a β-particle at each stage of the decay sequence.

What is the number of stages in the decay sequence?

A 20

B 16

C 8

D 4

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In β^- decay, a neutron decays to become a proton, β^- particle $^{0}_{-1}\beta$ and antineutrino v_e
  - $^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- An α-particle is made of two protons and two neutrons, $^{4}_{2}\alpha$
  - $^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- The difference in mass between
  - $^{222}_{86}Rn$ and $^{206}_{82}Pb$ is 222 – 206 = 16
- As beta decay does not decrease mass, the change must be due to alpha decay
  - Number of alpha particles released $= \frac{16}{4} = 4$
- The difference in atomic number
  - $^{222}_{86}Rn$ and $^{206}_{82}Pb$ is 86 – 82 = 4
  - The four alpha particles released would have emitted 4 × 2 = 8 protons
- The number of beta particles released must, therefore, be 4 to increase the proton number to 82

- Therefore, the total number of stages = 8

Question 19 Medium [1 mark]

An element emits an alpha particle from its radioactive nucleus.

The daughter nucleus then emits a beta particle, and then the daughter nucleus of that reaction emits another beta particle.

Which statement describes the final nuclide that is formed ?

A it is a nuclide of the same element but with different proton number

B it is a nuclide of a different element of higher proton number

C it is a different isotope of the original element

D it is identical to the original nuclide

Mark Scheme & Explanation:

*The correct answer is **C because:***

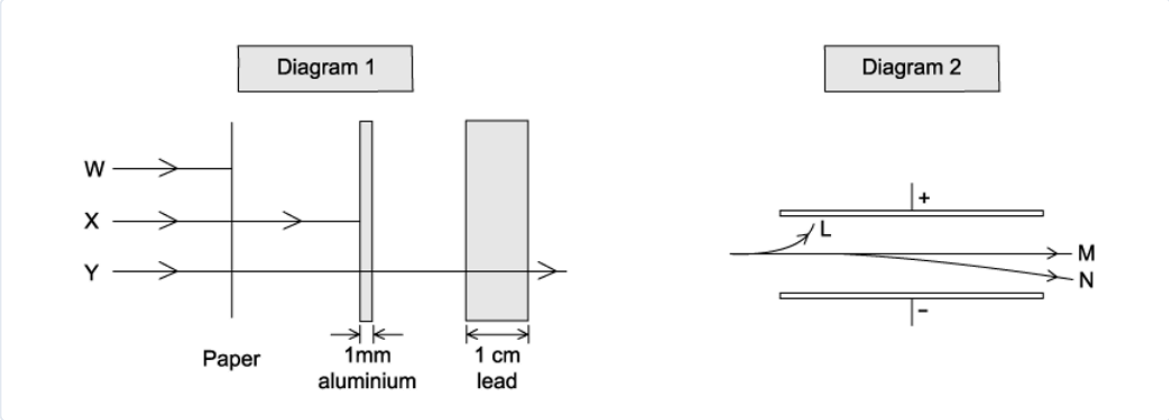
- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- The first stage emits an α-particle reducing the proton number by 2
- The next two stages both emit a β increasing the proton number by 2
  - The final nucleus must have the same number of protons and a different number of neutrons, therefore, option **C** is correct

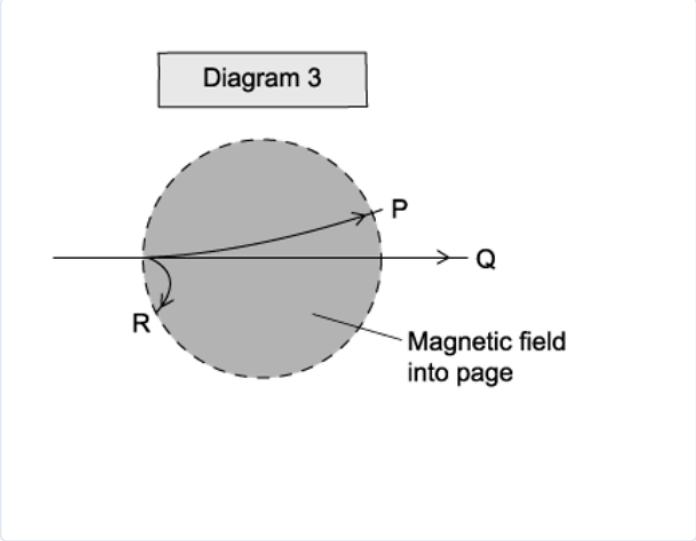
Question 20 Medium [1 mark]

Alpha, beta and gamma radiations:

1	are absorbed to different extents in solids
2	behave differently in an electric field
3	behave differently in a magnetic field

Diagrams 1, 2 and 3 illustrate these behaviours .





Which three labels on these diagrams refer to the **same** kind of radiation?

A X, L, R

B W. L, R

C W, L, P

D Y, M, P

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	−1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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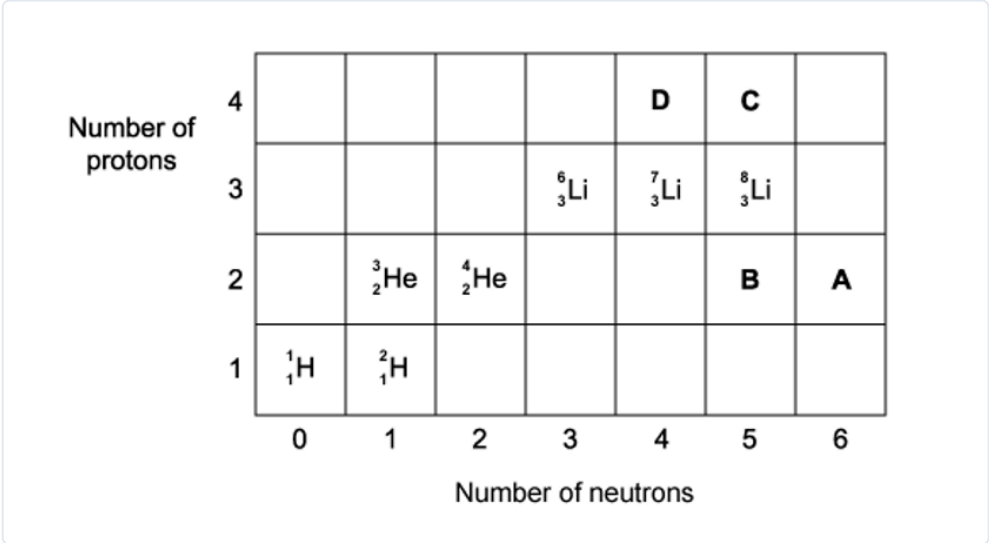
- Diagram 1 shows the absorbance of the radiation by solids:
  - W is an alpha particle stopped by paper
  - X is a beta particle stopped by 1 mm of aluminium
  - Y is a gamma-ray reduced by lead
- Diagram 2 shows the radiations passing an electric field:
  - L is a beta-particle attracted to the positive terminal
  - M is a gamma-ray not deflected by the electric field
  - N is an alpha particle attracted to the negative terminal
- Diagram 3 shows:
  - R is a beta particle deflected by the magnetic field
  - P is an alpha particle deflected by the magnetic field
  - Q is a gamma-ray is not deflected by the magnetic field
- The correct option is therefore **A**

Question 21 Medium [1 mark]

The nuclides shown in the grid below are arranged according to the number of protons and neutrons in each.

A nucleus of the nuclide $^{8}_{3}Li$ decays by emitting a β-particle.

What is the resulting nuclide ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- In β^- decay, a neutron decays to become a proton, β^- particle $^{0}_{-1}\beta$ and antineutrino v_e
  - $^{A}_{Z}X \rightarrow {}_{Z+1}{}^{A}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- This type of decay would result $^{8}_{3}Li$ in having one extra proton and one less neutron, so, the only correct option is **D**

Question 22 Medium [1 mark]

A thorium isotope has a nucleon number of 232 and a proton number of 90. It decays to form another isotope of with a nucleon number of 228.

How many alpha particles and beta particles are emitted during this decay?

	Alpha particles	Beta particles
A	0	4
B	1	2
C	1	1
D	2	1

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The question states that after decay it is an isotope of thorium which has a proton number of 90
- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 232 – 228 = 4, so, two neutrons and two protons are emitted
  - This would result in a proton number of 88, this is not correct as the final isotope should also have a proton number of 90
  - Therefore, two beta particles must be emitted to make up the final proton number
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
  - So, this would result in a proton number of 90

Therefore, one alpha particle and two beta particles are emitted

A is incorrect because in this option the final proton number would be 94

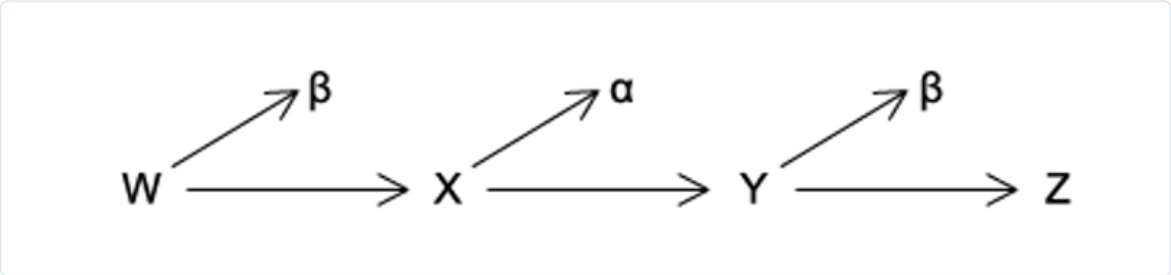
C is incorrect because in this option the final proton number would be 89

D is incorrect because in this option the final proton number would be 87

Question 23 Medium [1 mark]

Three successive radioactive decays are shown in the diagram below; each one results in a particle being emitted.

The first decay results in the emission of a β-particle. The second decay results in the emission of an α-particle. The third decay results in the emission of another β-particle.



Nuclides W and Z are compared. Which statement is correct ?

A W and Z are isotopes of the same element

B Z is a different element of reduced mass

C Z is a different element of lower atomic number

D W and Z are identical in all respects

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
  - The first stage emits an β-particle increasing the proton number by 1
  - The next stage emits an α-particle decreasing the proton number by 2
  - The final stage emits an β-particle increasing the proton number by 1
- The number of protons in Z are the same as W, but with a different number of neutrons

- Therefore, W and Z are isotopes of each other

Question 24 Medium [1 mark]

Antimatter is a particle that is an antiparticle to the corresponding particle. A positron is the antiparticle of an electron.

What is the difference between a positron and an electron ?

A Mass

B magnitude of charge

C Charge

D Spin

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- An antiparticle is a subatomic particle having the same mass as one of the particles of ordinary matter but opposite electric charge and magnetic movement
- A positron is the antiparticle of the negatively charged electron
- They only differ in their charge
  - The positron has a + ve charge and the electron a - ve charge

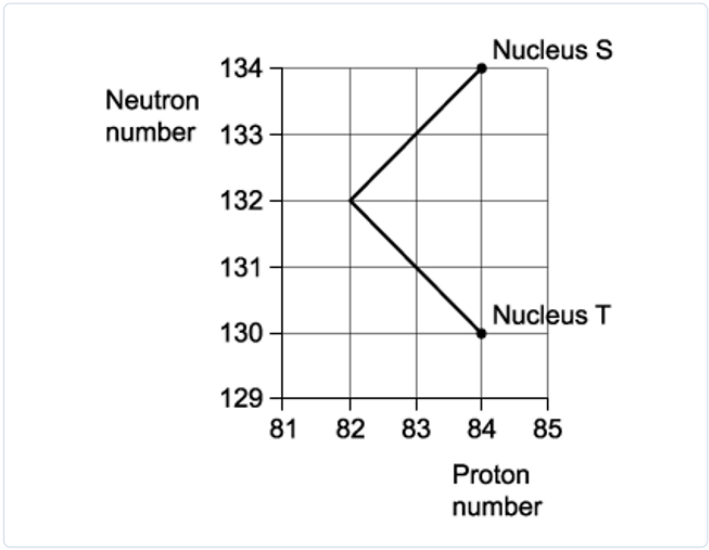
A is incorrect because the mass of an electron and positron are the same

B is incorrect because the magnitude of charge is the same; the positron has a charge of +1 and the electron -1

D is incorrect because the positron and electron both have the same spin of ½

Question 25 Medium [1 mark]

A sequence of radioactive decays is shown in the graph of neutron number against proton number.



Nucleus S is at the start of the sequence and, after the decays have occurred, nucleus T is formed.

What is emitted during the sequence of decays?

A one α-particle followed by one β-particle

B two β-particles followed by one α-particle

C two α-particles followed by two β-particles

D one α-particle followed by two β-particles

Mark Scheme & Explanation:

*The correct answer is **D because:***

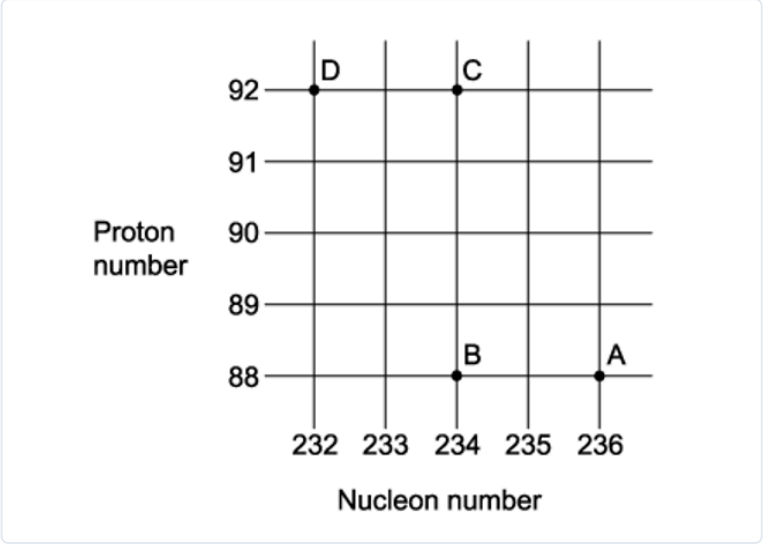
- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
  - Nucleus S goes from ${}^{134}_{84}S$ to ${}^{132}_{82}X$, so the first stage is α decay
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$

- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
    - Nucleus X goes from ${}^{132}_{82}X$ to ${}^{130}_{84}T$, so two β^- particles must have been released

Question 26 Medium [1 mark]

A radioactive substance with a nucleon number of 234 and a proton number of 90, decays by β-emission into a daughter product which in turn decays by further β-emission into a granddaughter product.

Which letter in the diagram represents the granddaughter product ?



A

B

C

D

Mark Scheme & Explanation:

The correct answer is ***C because:***

- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
  - Nucleus X goes from ${}^{234}_{90}X$ and undergoes two β^- decays
  - The nucleon number will not change, so, it will still be 234
  - The proton number will increase by two, so, it will be 92
- Therefore, option **C** correct

Question 27 Medium [1 mark]

Astatine is a radioactive substance; it has a nucleon number of 218 and a proton number of 85. When it decays it forms a polonium nucleus, emitting a β^- particle and an α-particle.

What are the nucleon number and the proton number of the polonium nucleus?

	Nucleon number	Proton number
A	214	84
B	214	83
C	216	83
D	215	82

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A because:***

- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- Astatine decays to form polonium emitting an ${}^{0}_{-1}\beta$ and ${}^{4}_{2}\alpha$
  - Stage 1 : ${}^{218}_{85}At \rightarrow {}^{218}_{86}X + {}^{0}_{-1}\beta$
  - Stage 2 : ${}^{218}_{86}X \rightarrow {}^{214}_{84}Po + {}^{4}_{2}\alpha$

Question 28 Medium [1 mark]

An element with an unstable nucleus decays by emitting an alpha particle to become the nucleus of a different element.

The nucleus of the new element is unstable and will emit either an α-particle or a β^- particle. This process continues until an isotope of the original element is formed.

What is the minimum possible number of the particles emitted ?

A 5

B 4

C 3

D 2

Mark Scheme & Explanation:

*The correct answer is **C because:***

- An α-particle is made of two protons and two neutrons, ${}^{4}_{2}\alpha$
  - ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle ${}^{0}_{-1}\beta$ and antineutrino v_e
  - ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + v_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
  - If the first decay emits an β-particle, it will increase the proton number by 1
  - If the second decay emits an α-particle, it will decrease the proton number by 2
  - If the third stage emits an β-particle, it will increase the proton number by 1
- The number of protons will remain the same after three decays, 1 α decay and 2 β decays
  - This makes them isotopes of each other

Question 29 Medium [1 mark]

When α-particles are directed at gold leaf

1	almost all α-particles pass through without deflection,
2	a few α-particles deviate through large angles.

What are the reasons for these effects?

	1	2
A	the positive charge in an atom is not concentrated enough to deflect an α-particle	occasionally an α-particle experiences many small deflections in the same direction
B	the gold nucleus is very small so most α-particles miss all nuclei	occasionally the path of an α-particle is close to a nucleus
C	most α-particles miss all gold atoms	a few α-particles bounce off gold atoms
D	most α-particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α-particles bounce back from the gold surface

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The gold foil experiment directed parallel beams of α-particles at a piece of gold foil only 10^{-6} m thick
- Most of the α-particles went straight through the foil
  - This is due to the small size of the nucleus of the atom
- Some of the α-particles were deflected through large angles
  - This is due to the path occasionally being close to the nucleus

A is incorrect because alpha particles are deflected by the positive charge when it is close enough to the nucleus

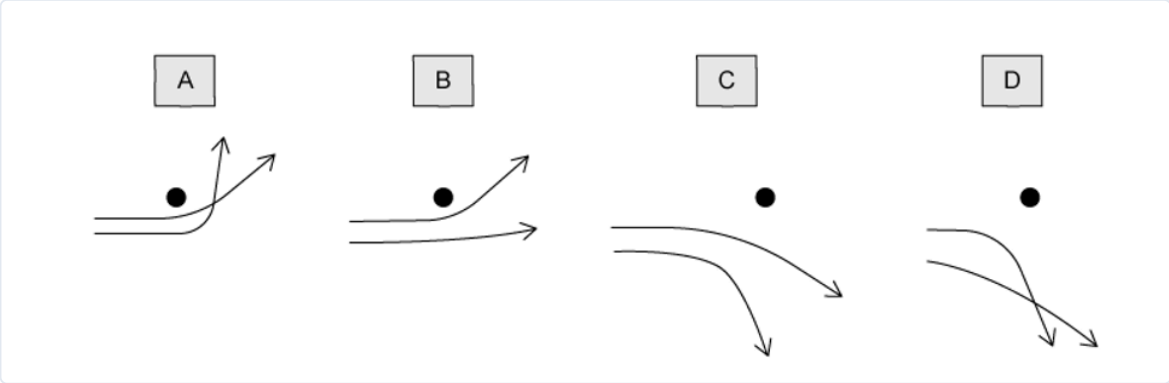
C is incorrect because the alpha particles did not bounce off the nuclei but were deflected due to repulsion from positively charged protons

D is incorrect because the gold foil used was only a few atoms thick to reduce the density of the particles

Question 30 Medium [1 mark]

Two α-particles with equal energies are deflected by a gold nucleus.

Which diagram best represents their paths ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The gold nucleus will cause a repulsive force on the alpha particle as they are all positively charged
- The gold nucleus would deflect the alpha particle away.
- The closer the particle the stronger the force and therefore the deflection
- The paths will not stay parallel due to a greater repulsive force on some of the particles

A is incorrect because the diagram shows an attraction towards the gold nucleus

B is incorrect because the diagram shows the paths not being crossed by the greater repulsive force, and an attraction of the alpha particle towards the nucleus

C is incorrect because the diagram shows the paths not being crossed by the greater repulsive force

Question 31 Hard [1 mark]

A stationary decay $^{210}_{84}Po$ of by α-emission has a total kinetic energy of E_k

$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \alpha$$

What is the kinetic energy of the daughter nucleus $^{206}_{82}Pb$?

A 0

B between 0 and $\frac{E_k}{2}$

C between $\frac{E_k}{2}$ and E_k

D E_k

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- By the Principle of Conservation of Momentum :
- Total momentum before α-emission = Total momentum after α-emission
  - Total momentum before α-emission = 0
  - Total momentum after α-emission = MV - mv
- Where *M* and *V* are the mass and velocity of $^{206}_{82}Pb$ and *m* and *v* are the mass and velocity of the α-particle
  - $0 = MV$ - mv
  - $MV = mv$
  - $M^2V^2 = m^2v^2$
  - $M\frac{1}{2}MV^2 = m\frac{1}{2}mv^2$
- Therefore, $\frac{KE\ of\ Pb\text{-}206}{KE\ of\ \alpha} = \frac{mass\ of\ \alpha}{mass\ of\ Pb\text{-}206}$
  - $\frac{KEPb}{KE\alpha} = \frac{4}{206} = 0.0194$
- From Conservation of Energy : $KEPb + KE\alpha = E_k$
  - $KE\alpha = \frac{KEPb}{0.0194}$
  - $KEPb + \frac{KEPb}{0.0194} = E_k$
  - $KEPb1 + \frac{1}{0.0194} = E_k$
  - So, $KEPb = \frac{E_k}{1 + \frac{1}{0.0194}} = 0.0190\ \ E_k$

- Therefore, the kinetic energy of $^{206}_{82}Pb$ is between 0 and $\frac{E_k}{2}$

Question 32 Hard [1 mark]

Which of the following equations correctly shows an α-particle causing a nuclear reaction ?

A $^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}n$

B $^{17}_{8}O + {}^{4}_{2}He \rightarrow {}^{20}_{9}F + {}^{1}_{1}P$

C $^{17}_{8}O + {}^{0}_{-1}e \rightarrow {}^{13}_{5}B + {}^{4}_{2}He$

D $^{14}_{7}N + {}^{1}_{1}p \rightarrow {}^{11}_{6}C + {}^{4}_{2}He$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An α-particle is a helium nucleus, which is represented by $^{4}_{2}He$
- A neutron is represented by $^{1}_{0}n$
- Since the α-particle is supposed to cause the reaction, it should appear on the left of the equation as a reactant
    - Hence, only option **A** or **B** can be correct
- In option **A**, the symbol of the neutron is wrongly represented as $^{1}_{1}n$
    - Therefore, since option **A** is not possible, this leaves only option **B**
- We can confirm this by comparing the nucleon numbers on each side
    - LHS: 17 + 4 = 21
    - RHS: 20 + 1 = 21
- Nucleon number is conserved, so option **B** is correct

A is incorrect because the symbol of the neutron is wrongly represented as $^{1}_{1}n$

C & D are incorrect because the α-particle causes the reaction, so, it should appear on the left of the equation as a reactant

Question 33 Hard [1 mark]

Thorium $^{232}_{90}Th$ decays through a series of transformations. The particles emitted in successive transformations are

$$\alpha \quad \beta \quad \beta \quad \gamma \quad \alpha$$

The resulting nuclide may be represented by

A $^{230}_{82}Pb$

B $^{227}_{85}At$

C $^{225}_{87}Fr$

D $^{224}_{88}Ra$

Mark Scheme & Explanation:

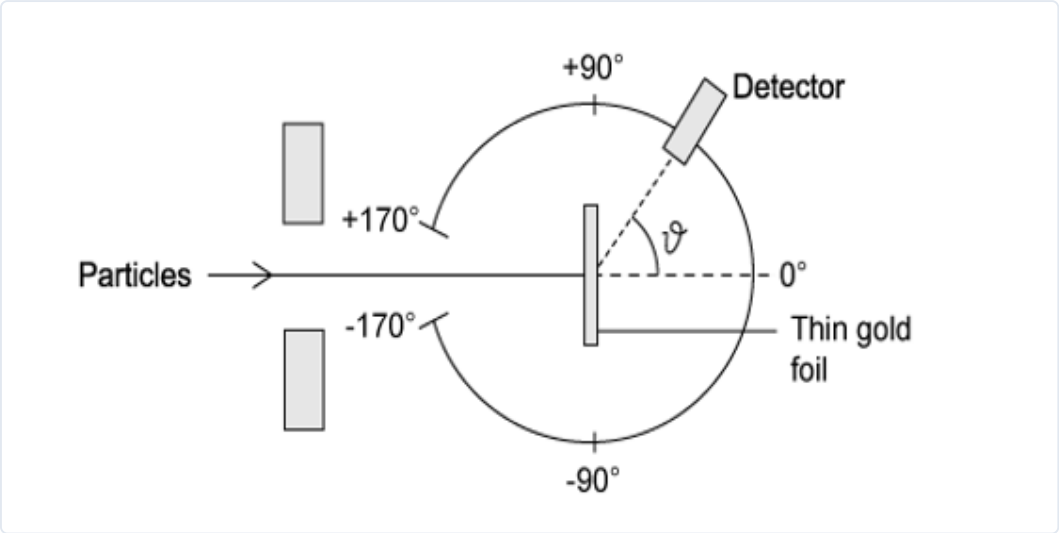
*The correct answer is **D** because:*

- During α-decay, a helium nucleus $^{4}_{2}He$ is released from each starting atom
  - The mass number decreases by 4 and the proton number by 2
- During β radiation, an electron (or positron) is released
  - There is a gain of 1 proton because a neutron splits to form a proton and electron
  - There is no change in atomic number
- During γ radiation, γ rays are emitted but there is no change in atomic or proton number
- We can draw a flow diagram of the processes involved:

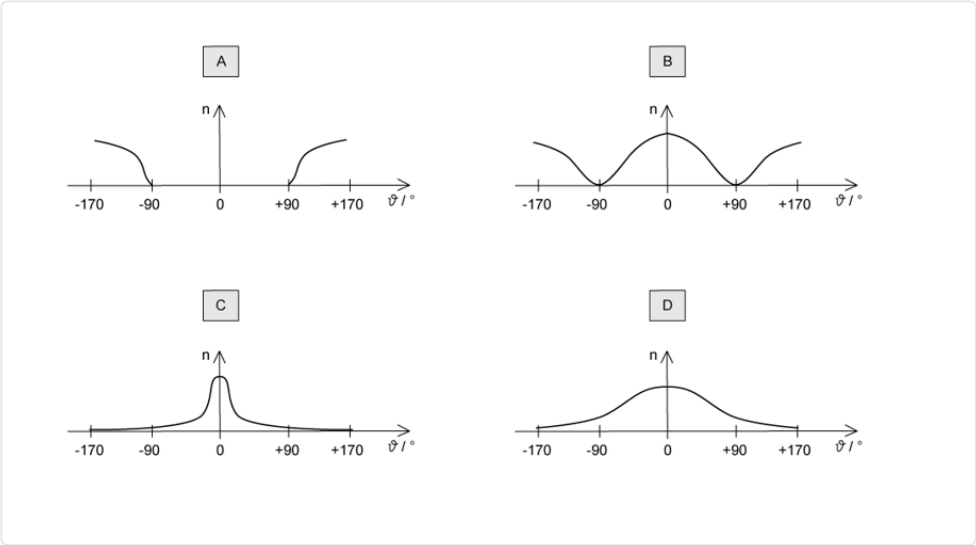
  $^{232}_{90}Th \xrightarrow{\alpha} {}^{228}_{88}Ra \xrightarrow{\beta} {}^{228}_{89}Ac \xrightarrow{\beta} {}^{228}_{90}Th \xrightarrow{\gamma} {}^{228}_{90}Th \xrightarrow{\alpha} {}^{224}_{88}Ra$
- Therefore, the end nuclide is $^{224}_{88}Ra$

Question 34 Hard [1 mark]

In an α-particle scattering experiment, a student set up the apparatus below to determine the number n of α-particles incident per unit time on a detector held at various angles θ.



Which of the following graph best represents the variation of n with θ ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The gold foil experiment directed parallel beams of α-particles at a piece of gold foil only 10^{-6} m thick

- The observations were as follows:
    - Most of the α-particles went straight through the foil - so, the largest value of n is at small angles
    - Some of the α-particles were deflected slightly - here, n drops quickly with increasing angle of deflection θ
    - Some α-particles, about 1 in 20 000, were deflected by more than 90˚, appearing to bounce back off the foil - so, the value of n is very small at large angles
- Therefore, the graph that matches this description is **C**

A & B are incorrect because these graphs show the majority of α-particles were deflected through large angles which is the opposite of what was observed

D is incorrect because this graph shows many particles were deflected through a variety of angles, however, the observations show that the majority of particles passed straight through the foil with little to no deflection

Question 35 Hard [1 mark]

An unknown isotope ${}^{A}_{Z}X$ decays through a series of transformations into Thorium-228. The atomic number of thorium is 90.

The successive transformations are:

$$\gamma \quad \beta^{-} \quad \alpha \quad \alpha \quad \gamma$$

Which row states the correct isotope name, atomic number and mass number of the unknown isotope ${}^{A}_{Z}X$?

	Isotope name	Atomic number / Z	Mass number / A
A.	Neptunium	93	235
B.	Plutonium	94	237
C.	Neptunium	93	236
D.	Uranium	92	232

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When α-decay occurs
  - A helium ${}^{4}_{2}He$ particle is emitted
  - The atomic number decreases by two and the mass number by four
- When β^{-} -decay occurs
  - A neutron transforms into a proton and an electron
  - The mass number remains the same but the atomic number increases by one
- When γ-decay occurs
  - A γ ray is emitted so there is no change in the mass or atomic number
- The end nuclide is Thorium-228

- The isotope can be written as $^{228}_{90}Th$

- The flow diagram for the decay of $^{A}_{Z}X$is

$$^{236}_{93}Np \xrightarrow{\gamma} {}^{236}_{93}Np \xrightarrow{\beta^-} {}^{236}_{94}Pu \xrightarrow{\alpha} {}^{232}_{92}U \xrightarrow{\alpha} {}^{228}_{90}Th \xrightarrow{\gamma} {}^{228}_{90}Th$$

- Therefore the original isotope is Neptunium-236
- This is option **C**
- In order to find the original unknown isotope, you will need to work backwards from $^{228}_{90}Th$
  - For α-decay add four onto the mass number and two onto the atomic number
  - For β^--decay minus one from the atomic number and keep the mass number the same